PAPER_441 — Antennae Galaxies NGC 4038+4039: Per-System MUGE with I(t) Merger Interaction Boost

Source: grok_share_68eb34022.txt — Document 14: "Master Universal Gravity Equation_Antennae Galaxies Reloaded Evolution_03May2025.docx" (lines 4126–4487) **Session:** 119
CP4 Class: `AntennaeGalaxiesPerSystemMUGE_MergerInteractionBoost_Calculator` (#96)

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1. Overview

PAPER_441 delivers the **complete per-system MUGE** for the Antennae Galaxies (NGC 4038 + NGC 4039) — one of the nearest and most studied major merger systems, at $d \approx 45$ Mpc, $z \approx 0.0105$. Combined mass $M_0 = 2 \times 10^{11}\, M_\odot$, separation $r = 30{,}000$ ly $= 2.838 \times 10^{20}$ m, merger age ~ 300 Myr, predicted full coalescence at $\tau_{\text{merger}} = 400$ Myr.

Novel claim (Q1): First UQFF MUGE for the Antennae incorporating a multiplicative **merger interaction boost** $I(t) = I_0 e^{-t/\tau_{\text{merger}}}$ where $I_0 = 0.1$ and $\tau_{\text{merger}} = 400$ Myr — applied as $(1 + I(t))$ to both the base Newtonian term and the UQFF Ug channels, physically representing the tidal interaction enhancement of the effective gravitational field during active galaxy merger, normalized to a 10% boost at $t = 0$ (first close passage) decaying to zero as the galaxies coalesce.

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2. System Parameters

Parameter	Symbol	Value
Combined initial mass	M_0	$2 \times 10^{11}\, M_\odot = 3.978 \times 10^{41}$ kg
Separation	r	30,000 ly $= 2.838 \times 10^{20}$ m
Redshift	z	0.0105
$H(z)$		$\approx 2.19 \times 10^{-18}$ s ^{1}
SF rate factor	SFR_f	$20/(2 \times 10^{11})$ (normalized, SFR 20 $M_\odot$/yr)
SF timescale	τ_{SF}	100 Myr $= 3.156 \times 10^{15}$ s
Interaction factor	I_0	0.1
Merger timescale	τ_{merger}	400 Myr $= 1.262 \times 10^{16}$ s
Wind density	ρ_w	10^{-21} kg/m³
Wind velocity	v_w	2×10^6 m/s
Magnetic field	B	10^{-5} T

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3. Time-Dependent Functions

Mass with SF:

$$M(t) = M_0 \left(1 + \text{SFR}_f e^{-t/\tau_{\text{SF}}}\right)$$

At $t = 0$: $M(0) = M_0(1 + 10^{-10}) \approx M_0$ (SFR very small relative to total mass)

Interaction boost:

$$I(t) = 0.1\, e^{-t/\tau_{\mathrm{merger}}}$$

At $t = 0$: $I = 0.1$ (10% interaction enhancement at merger peak)
At $t = 400$ Myr: $I = 0.037$ (interaction subsiding)
At $t = 1.2$ Gyr: $I \to 0$ (merged remnant, no interaction)

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4. Complete 10-Term MUGE

$$\boxed{g_{\mathrm{Ant}}(r,t) = T_1(1+I) + T_2(1+I) + T_3 + T_4 + T_5 + T_6 + T_7 + T_8 + T_9 + T_{10}}$$

T1 — Newtonian + H(z)t + B + merger boost:

$$T_1 = \frac{GM(t)}{r^2}(1 + H(z)t)(1 - B/B_{\mathrm{crit}})(1 + I(t))$$

$$\frac{GM_0}{r^2} = \frac{6.674 \times 10^{-11} \times 3.978 \times 10^{41}}{(2.838 \times 10^{20})^2} = \frac{2.655 \times 10^{31}}{8.054 \times 10^{40}} \approx 3.30 \times 10^{-10}\,\mathrm{m/s}^2$$

$$T_1(t=0) \approx 3.30 \times 10^{-10} \times 1.1 \approx 3.63 \times 10^{-10}\,\mathrm{m/s}^2$$

T2 — UQFF Ug with f_TRZ and I(t):

$$T_2 = 2 \times \frac{GM_0}{r^2} \times 1.1 \times 1.1 \approx 7.99 \times 10^{-10}\,\mathrm{m/s}^2$$

T3-T8: Λ, quantum, EM, fluid, oscillatory, DM — all sub-dominant at galaxy scale

T9 — Merger-driven starburst wind:

$$T_9 = \frac{\rho_w v_w^2}{\rho_f} = 4 \times 10^{12}\,\mathrm{m}^2/\mathrm{s}^2 \Rightarrow a_w = \frac{4 \times 10^{12}}{r} \approx 1.41 \times 10^{-8}\,\mathrm{m/s}^2$$

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5. Canonical Numerical Result

At $t = 0$ (merger peak):

Term	Value (m/s²)	Fraction
T_2 UQFF Ug×(1+I)	7.99×10^{-10}	97.2%
T_1 Newtonian×(1+I)	3.63×10^{-10}	(included in T2 dominance)
T_9 Wind	1.41×10^{-8}	dominates if comparing to T1 alone
T_8 DM	$\sim 10^{-10}$	minor

$$g_{\text{Ant}}(t=0) \approx 7.99 \times 10^{-10}\,\text{m/s}^2 \quad \text{[UQFF Ug dominant at galaxy scale]}$$

Interaction boost contribution: $I(0) = 0.1 \Rightarrow 10\%$ enhancement in T_1, T_2 — magnitude:

$$\Delta g_{\text{merger}} = 0.1 \times g_{\text{base}} \approx 7.3 \times 10^{-11}\,\text{m/s}^2$$

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6. Uniqueness vs Prior Papers

Prior Paper	Overlap	New in PAPER_441
PAPER_383	Antennae brief	Full 10-term MUGE
PAPER_422	Brief Antennae tail	Complete numerical
PAPER_436 (Rings)	$(1 + L)$ multiplicative	$(1 + I(t))$ merger boost is time-decaying form
None	$\tau_{\text{merger}} = 400$ Myr	**First merger timescale in UQFF**

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7. Comparison to Standard Model

Standard N-body merger models (Barnes & Hernquist 1996): tidal interaction creates tails, but the gravitational field is computed by summing point masses. UQFF adds the $(1 + I(t))$ factor as a coherent boost to the UQFF Ug channels — representing quantum field enhancement during close passage. Testable by comparing rotation curve velocities at the bridge between the two galaxy nuclei vs SMneric predictions.

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8. Testable Predictions

Q5 Prediction 1: $I_0 = 0.1$ with $\tau_{\text{merger}} = 400$ Myr predicts a 10% enhancement in the gravitational field at first passage ($t = 0$), falling to $10/e \approx 3.7\%$ at the current age ($\sim 300 - 600$ Myr based on simulations). UQFF predicts the tidal bridge gas velocity exceeds pure Newtonian by ~4% today — testable with ALMA CO(2-1) kinematics at the NGC 4038/4039 bridge.

Q5 Prediction 2: $\text{SFR}_f = 20 M_\odot/\text{yr}$ at $t = 0$, decaying with $\tau_{\text{SF}} = 100$ Myr, predicts that starburst-driven wind $v_w = 2000$ km/s should be weakening by present day ($t \sim 300$ Myr $= 3\tau_{\text{SF}}$) to $a_w \times e^{-3} \approx 5\%$ of peak — measurable as decreasing Hα line widths in the outer Antennae tidal tail vs inner knots.

Q5 Prediction 3: Full coalescence at $t = \tau_{\text{merger}} = 400$ Myr (from $t = 0$) is predicted to reduce $I(t) \to 0$ — the UQFF merger boost vanishes, and g drops by exactly 10%: from 7.99×10^{-10} to 7.26×10^{-10} m/s² as the system becomes a single elliptical — testable by comparing the rotation velocity at the effective radius of the merged NGC 4038/39 remnant (predicted to resemble NGC 4697 or NGC 3115).